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Administering a lymphocytic choriomeningitis virus (LCMV) with improved tumor regression properties

Abstract

The invention relates to arenaviruses for use in the treatment and/or prevention of tumors and also a method for preparing arenaviruses with (improved) tumor-regressive properties.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 16/267,095, filed Feb. 4, 2019, now U.S. Pat. No. 11,801,294, which is a divisional application of U.S. patent application Ser. No. 15/567,343, filed Oct. 17, 2017, which is the United States National stage patent application of International Application No. PCT/EP2016/058347, filed Apr. 15, 2016, which claims the right of priority of German patent application No. 102015207036.0 filed Apr. 17, 2015, with the German Patent Office, the entire contents of each of which are incorporated herein for all purposes.

SEQUENCE LISTING

(1) The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on Mar. 27, 2024, is named "SCH-5000-CT1_SL.xml" and is 2,758, 175 bytes in size. The sequence listing

contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(2) The invention relates to arenaviruses for use in the treatment and/or prevention of tumors and also methods for preparing arenaviruses with (improved) tumor-regressive properties.

(3) Arenaviruses belong to the family of human pathogenic, pleomorphic RNA viruses. Diseases with these viruses belong to the zoonoses due to their natural reservoir in animals, predominantly rodents. Zoonoses refer to diseases that can be transferred from the animal to humans and vice versa from humans to the animal.

(4) At least eight arenaviruses are known to cause illness in humans. Typical are aseptic meningitis and haemorrhagic fever. Known viruses which can trigger a disease in humans are the lymphocytic choriomeningitis virus (LCMV), Guanarito virus (GTOV), Junin virus (JUNV), Lassa virus (LASV), Lujo virus (LUJV) Machupo virus (MACV), Sabia virus (SABV) and the Whitewater Arroyo virus (WWAV).

(5) Arenaviruses are generally divided into two groups, namely the Old World arenaviruses and the New World arenaviruses. These groups differ geographically and genetically. Old World arenaviruses, such as the lymphocytic choriomeningitis virus, have been found in countries of the eastern hemisphere, such as European, Asian and African countries. In contrast, New World arenaviruses have been found in countries of the western hemisphere, such as Argentina, Bolivia, Venezuela, Brazil and the United States of America.

(6) The name of the virus family is derived from the Latin arenosus (sandy) and arena (sand) to describe the sandy ribosomal structure within the virions. The virions of the arenaviruses have a round to irregular shape and have a diameter, depending on species and preparation of the test material, from 50 nm to 300 nm, usually between 110 nm and 130 nm. Club-shaped glycoprotein spikes, 8 nm to 10 nm long, are embedded in the virus envelope. The individual spikes consist of a tetramer of the viral envelope protein.

(7) The virions also comprise two closed-ring capsids with helical symmetry. The length of the capsids varies between 450 nm and 1300 nm. One molecule of the viral RNA (ribonucleic acid) polymerase (L-protein) is attached to each of them.

(8) Each capsid comprises one molecule of a single-stranded RNA with mixed (i.e. ambisense, +/-) polarity. The two single-stranded RNA molecules represent the viral genome. They are referred to as L (large) and S (small) and are about 7.5 kb (kilobases) or 3.5 kb (kilobases) large. Although the capsids are closed ring-shaped, the RNA strands are linear and thus not circular. A 19 to 30 base long sequence at the 3' end of the RNA is present on both strands and is also conserved within the virus family.

(9) Very exceptional morphologically is the presence of an alternating number of cellular ribosomes within the virions, which give the viral particles their "sandy" appearance. Similarly, in purified virus preparations, a number of different cellular RNAs (also including ribosomal RNA) and also replicative forms of viral RNA are found, as are diverse viral mRNAs (messenger ribonucleic acids bound to the ribosomes) and complete complementary strands of the virus genome. These non-genomic RNAs are found in varying amounts all lying outside the abovementioned capsids.

(10) The use of arenaviruses as vaccination vectors is known. A prominent example is the vaccination virus Candid #1 used against Argentinian hemorrhagic fever. This is a vaccination variant of the Junin virus.

(11) Known from WO 2009/083210 A1 is the use of replication defects, i.e. genetically modified arenavirus particles (virions), inter alia, for the treatment of neo-plastic diseases such as, for example, melanoma, prostate carcinoma, breast carcinoma and lung carcinoma. In the publication "Development of replication-defective lymphocytic choriomeningitis virus vectors for the induction of potent CD8.sup.+ T cell immunity" (Nature Medicine, Vol. 16, No. 3, March 2010, pp.

339-345; doi: 10.1038/Nm.2104), cancer immunotherapy is mentioned as a potential area of application for such viral particles.

(12) Furthermore, WO 2006/008074 A1 discloses the use of packaging cells, which produce retroviral virions pseudotyped with arenavirus glycoprotein, for gene therapy of solid tumors.

(13) The methods for the treatment of tumors described in the prior art are based on the use of virus particles which are very complicated to produce by genetic engineering. In the case of gene therapy treatment methods, it is frequently not possible to achieve adequate, therapeutically effective transduction of the tumor tissue with genetically engineered virions or packaging cells which produce virions.

SUMMARY OF THE INVENTION

Object and Solution

(14) The present invention is therefore based on the object of providing a simpler and, in particular, more efficient therapeutic solution for tumors, in particular carcinomas and sarcomas, compared to the prior art.

(15) This object is achieved according to the invention by an arenavirus according to independent claim **1**, by a medicament according to claim **14** and also by an in vitro method according to independent claim **15**. Preferred embodiments are defined in the dependent claims. The wording of all claims is hereby incorporated by reference into the content of the description. An additional subject of the invention, which achieves the object of the invention, is disclosed in the description.

(16) According to a first aspect, the invention relates to an arenavirus for use in the treatment and/or prevention of a tumor, preferably a malignant tumor, in humans or animals.

(17) The arenavirus is preferably characterized in that it is free of genomic foreign RNA, i.e. it does not comprise any genomic foreign RNA. In other words, the genome of the arenavirus is preferably free of foreign RNA or preferably comprises no foreign RNA.

(18) In the context of the present invention, the expression “genomic foreign RNA” is intended to mean an RNA (ribonucleic acid) or RNA sequence which does not occur or is not present in the genome of a wild-type arenavirus or in the genome of a mutant of a wild-type arenavirus (mutated arenavirus), in particular in the genome of a natural mutant of a wild-type arenavirus (naturally mutated arenavirus). Examples of foreign RNA are artificial or synthetic RNA molecules, RNA of organisms and RNA from other viruses.

(19) In the context of the present invention, (in accordance with the understanding of those skilled in the art), the expression “wild-type arenavirus” is understood to mean an arenavirus whose genome is the genetically normal form occurring in nature.

(20) In the context of the present invention, (in accordance with the understanding of those skilled in the art), the expression “mutant of a wild-type arenavirus” or “mutated arenavirus” is understood to mean an arenavirus whose genome comprises a spontaneous, i.e. naturally-induced, modification or modification induced by mutagenesis, compared to the wild-type genome.

(21) Accordingly, in the context of the present invention, (in accordance with the understanding of those skilled in the art), the expression “natural mutant of a wild-type arenavirus” or “naturally mutated arenavirus” is understood to mean an arenavirus whose genome comprises a spontaneous, i.e. naturally-induced, modification, compared to the wild-type genome. A naturally mutated arenavirus can be produced preferably by passage, in particular serial passage, which will be discussed in more detail below.

(22) The invention is based on the surprising finding that arenaviruses without genomic foreign RNA are able to effect tumor regression. Tumor regression is due to an activation or stimulation of congenital and adaptive immune cells caused by the arenaviruses. The activated immune cells secrete increased antitumoral cytokines such as interferon- α and interferon- γ , thereby counteracting or repelling the tumor. A further surprising finding is the realization that the arenaviruses cause a significantly increased secretion of antitumoral cytokines in the case of a tumor manifestation. Arenaviruses without genomic foreign RNA are thus suitable for use in tumor treatment. This has

been successfully verified by the applicant by means of animal experiments. For this purpose, mice were used, inter alia, in which growth of human tumors is possible.

(23) In a further embodiment, the arenavirus is also free of non-genomic foreign RNA. In the context of the present invention, the expression “non-genomic foreign RNA” is intended to mean an RNA or RNA sequence, apart from the arenavirus genome, which does not occur or is not present in a wild-type arenavirus or a mutant of a wild-type arenavirus (mutated arenavirus), in particular of a natural mutant of a wild-type arenavirus (naturally mutated arenavirus). In other words, in this embodiment the arenavirus does not comprise overall any foreign RNA, i.e. neither genomic foreign RNA nor non-genomic foreign RNA.

(24) In a preferred embodiment, the arenavirus is a wild-type arenavirus.

(25) In a further embodiment, the arenavirus is a natural mutant of a wild-type arenavirus, i.e. a naturally mutated arenavirus.

(26) The natural mutant or the naturally mutated arenavirus is preferably produced by passage, in particular multiple passage, in host animals and/or host cells.

(27) The natural mutant or the naturally mutated arenavirus is particularly preferably produced by serial passage in host animals and/or host cells.

(28) The arenavirus provided according to the invention is thus preferably an arenavirus which is produced starting from its wild-type form by passage, preferably serial passage, in host animals and/or host cells.

(29) The host animals mentioned in the preceding paragraphs are preferably rodents, particularly mice. The host cells mentioned in the previous paragraphs, on the other hand, are preferably dendritic cells or tumor cells.

(30) In the context of the present invention, in accordance with the understanding of those skilled in the art, the term “passage” is understood to mean a multiple, regular introduction of the arenavirus into host animals and/or host cells. In the context of the present invention, in accordance with the understanding of those skilled in the art, the expression “serial passage” is understood to mean a multiple, regular introduction of the arenavirus into different host animals, preferably of the same type, and/or different cells, preferably of the same type. Due to the multiple changes of environment (host animal and/or host cell), the arenavirus is subject to an increased adaptation pressure or mutational pressure, thereby increasing the likelihood of advantageous mutations occurring in the genome of the arenavirus from the perspective of tumor regression.

(31) In a further embodiment, the tumor is selected from the group comprising or consisting of carcinoma, melanoma, blastoma, lymphoma and sarcoma.

(32) In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term “carcinoma” is intended to mean malignant neoplasia of epithelial origin.

(33) In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term “sarcoma” is intended to mean malignant neoplasia of mesodermal origin.

(34) In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term “melanoma” is intended to mean malignant neoplasia of melanocytic origin.

(35) In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term “lymphoma” is intended to mean malignant neoplasia of lymphocytic origin.

(36) In the context of the present invention, (in accordance with the understanding of those skilled in the art), the term “blastoma” is intended to mean malignant neoplasia of embryonic origin.

(37) In a preferred embodiment, the carcinoma is selected from the group comprising or consisting of anal carcinoma, bronchial carcinoma, lung carcinoma, endometrial carcinoma, gallbladder carcinoma, hepatocellular carcinoma, testicular carcinoma, colorectal carcinoma, laryngeal carcinoma, oesophageal cancer, gastric carcinoma, breast carcinoma, renal carcinoma, ovarian carcinoma, pancreas tumor, pharyngeal carcinoma, prostate carcinoma, thyroid carcinoma and cervical carcinoma.

(38) In a preferred embodiment, the sarcoma is selected from the group comprising or consisting of

angiosarcoma, chondrosarcoma, Ewing's sarcoma, fibrosarcoma, Kaposi's sarcoma, liposarcoma, leiomyosarcoma, malignant fibrous histiocytoma, neurogenic sarcoma, osteosarcoma and rhabdomyosarcoma.

(39) In a preferred embodiment, the arenavirus is an Old World arenavirus which is preferably selected from the group comprising or consisting of Ippy virus (IP-PYV), Lassa virus (LASV), lymphocytic choriomeningitis virus (LCMV), Mobala virus (MOBV) and Mopeia virus (MOPV).

(40) In a particularly preferred embodiment, the arenavirus is the lymphocytic choriomeningitis virus, preferably a strain which is selected from the group comprising or consisting of WE, Armstrong, Clone 13 and Docile.

(41) In a preferred embodiment, the arenavirus is a New World arenavirus, which is preferably selected from the group comprising or consisting of Allpaahuayo virus (ALLV), Amapari virus (AMAV), Bear Canyon virus (BCNV), Chapare virus, Cupixi virus (CPXV), Flexal virus (FLEV), Guanarito virus (GTOV), Junin virus (JUNV), Latino virus (LATV), Machupo virus (MACV), Oliveros virus (OLVV), Parana virus (PARV), Pichinde virus (PICV), Pirital virus (PIRV), Sabia virus (SABV), Tacaribe virus (TCRV), Tamiami virus (TAMV) and Whitewara Arroyo virus (WWAV).

(42) In a particularly preferred embodiment, the arenavirus is a Junin virus, in particular the strain Candid #1 (Candid No.1).

(43) In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an S-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 1 (according to sequence listing).

(44) In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an L-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 2 (according to sequence listing).

(45) In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an S-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 3 (according to sequence listing).

(46) In a further embodiment, the Junin virus, in particular the strain Candid #1 (Candid No.1), has a nucleic acid sequence, in particular an L-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 4 (according to sequence listing).

(47) In a further embodiment, the lymphocytic choriomeningitis virus (LCMV virus) mentioned above has a nucleic acid sequence, in particular an S-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 5 (according to sequence listing).

(48) In a further embodiment, the lymphocytic choriomeningitis virus (LCMV virus) mentioned above has a nucleic acid sequence, in particular an L-ribonucleic acid sequence or ambisense sequence, according to SEQ ID No. 6 (according to sequence listing).

(49) In a further embodiment, the arenavirus is isolated from tumor lysates, organ lysates, urine or blood.

(50) In an alternative embodiment, the arenavirus is isolated from a cell culture medium, in particular from a human tumor cell line.

(51) In a further embodiment, the arenavirus is for administration in the form of virions, i.e. in the form of arenavirus particles, which are outside a cell.

(52) In a further embodiment, the arenavirus is provided, preferably prepared, for local, in particular intramuscular, intraperitoneal or subcutaneous administration. In other words, the arenavirus is used in a further embodiment for local, in particular intramuscular, intraperitoneal or subcutaneous administration.

(53) Preferably the arenavirus is provided, preferably prepared, for local administration at a dose of 1 PFU (Plaque Forming Unit)/kg body weight to 10.sup.12 PFU/kg body weight, particularly 10.sup.2 PFU/kg body weight to 10.sup.6 PFU/kg body weight, preferably 10.sup.3 PFU/kg body weight to 10.sup.5 PFU/kg body weight. In other words, the arenavirus is used, preferably for local

administration, at a dose of 1 PFU (Plaque Forming Unit)/kg body weight to 10.sup.12 PFU/kg body weight, particularly 10.sup.2 PFU/kg body weight to 10.sup.6 PFU/kg body weight, preferably 10.sup.3 PFU/kg body weight to 10.sup.5 PFU/kg body weight.

(54) In an alternative embodiment, the arenavirus is provided, preferably prepared, for systemic, in particular intravenous, administration. In other words, the arenavirus is used in an alternative embodiment for systemic, in particular intravenous, administration.

(55) Preferably, the arenavirus is provided, preferably prepared, for systemic administration at a dose of 1 PFU/kg body weight to 10.sup.12 PFU/kg body weight, particularly 10.sup.2 PFU/kg body weight to 10.sup.6 PFU/kg body weight, preferably 10.sup.3 PFU/kg body weight to 10.sup.5 PFU/kg body weight. In other words, the arenavirus is used, preferably for systemic administration, at a dose of 1 PFU/kg body weight to 10.sup.12 PFU/kg body weight, particularly 10.sup.2 PFU/kg body weight to 10.sup.6 PFU/kg body weight, preferably 10.sup.3 PFU/kg body weight to 10.sup.5 PFU/kg body weight.

(56) According to a second aspect, the invention relates to a medicament for use in the treatment and/or prevention of a tumor, in particular a malignant tumor.

(57) The medicament is characterized in particular by the fact that it has an arenavirus according to the first aspect of the invention.

(58) The medicament preferably further comprises a pharmaceutically acceptable carrier. The carrier may in particular be selected from the group comprising or consisting of water, saline solution, buffer solution and cell culture medium.

(59) In a further embodiment, the medicament also comprises an active ingredient. The active ingredient can in particular be a cytostatic agent, an antibody and/or a cytokine.

(60) With regard to further features and advantages of the medicament, in particular of the arenavirus and also the tumor, reference is made fully to the present description in order to avoid repetitions.

(61) According to a third aspect, the invention relates to a method for producing an arenavirus with tumor-regressive, i.e. tumor-repelling/counteracting properties or improved tumor-regressive properties.

(62) The tumor is preferably a malignant tumor, preferably a carcinoma, melanoma, blastoma, lymphoma or sarcoma. Accordingly, the method is preferably a method for producing an arenavirus with (improved) carcinoma-, melanoma-, blastoma-, lymphoma- or sarcoma-regressive properties.

(63) The method comprises the following steps: a) infecting dendritic cells or tumor cells with an arenavirus, b) culturing the arenavirus in the infected dendritic cells or infected tumor cells and c) isolating the cultured arenavirus or a subset of the cultured arenavirus from the infected dendritic cells or infected tumor cells.

(64) The sequence of steps a) to c) may also be referred to as a (single) passage of the arenavirus in the dendritic cells or tumor cells.

(65) In a preferred embodiment, the arenavirus is an arenavirus which has been subjected to a serial passage in host animals prior to carrying out step a). For further details and advantages, reference may be made to the corresponding statements made in the context of the fourth aspect of the invention.

(66) In a further embodiment, the dendritic cells or tumor cells are in the form of a cell culture (dendritic cell culture or tumor cell culture).

(67) The tumor cells are preferably malignant tumor cells, in particular carcinoma, melanoma, blastoma, lymphoma or sarcoma cells.

(68) The carcinoma cells may be selected from the group comprising or consisting of anal carcinoma cells, bronchial carcinoma cells, lung carcinoma cells, endometrial carcinoma cells, gallbladder carcinoma cells, hepatocellular carcinoma cells, testicular carcinoma cells, colorectal carcinoma cells, laryngeal carcinoma cells, oesophageal carcinoma cells, gastric carcinoma cells, breast carcinoma cells, renal carcinoma cells, ovarian carcinoma cells, pancreas tumor cells,

pharyngeal carcinoma cells, prostate carcinoma cells, thyroid carcinoma cells and cervical carcinoma cells.

(69) The sarcoma cells may be selected from the group comprising or consisting of angiosarcoma cells, chondrosarcoma cells, Ewing's sarcoma cells, fibrosarcoma cells, Kaposi's sarcoma cells, liposarcoma cells, leiomyosarcoma cells, malignant fibrous histiocytoma cells, neurogenic sarcoma cells, osteosarcoma cells and rhabdo-myosarcoma cells.

(70) In a further embodiment, the tumor cells are immortalized immune cells, in particular immortalized macrophages.

(71) The dendritic cells or tumor cells are infected according to step a) preferably by adding the arenavirus to the cells.

(72) In a preferred embodiment, the sequence of steps a) to c) is repeated with new, in particular non-infected, dendritic cells, preferably of the same type, or with new, in particular non-infected, tumor cells, preferably of the same type (the same tumor type).

(73) In a particularly preferred embodiment, the sequence of steps a) to c) is repeated many times. By way of preference, new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are used for each repetition.

(74) The sequence of steps a) to c) is preferably repeated once to 1000 times, particularly 10 times to 100 times, preferably 30 times to 60 times, wherein new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are used for each repetition.

(75) By multiple repetition of steps a) to c), the arenavirus is constantly forced to adapt to a new environment, i.e. to new dendritic cells or tumor cells. This permanent adaptation pressure favors the occurrence of mutations, which can produce or improve the tumor-regressive properties of the arenavirus.

(76) During the culturing of the arenavirus according to step b), a replication of the arenavirus genome and a propagation of the arenavirus occur within the dendritic cells or tumor cells.

(77) The arenavirus according to step b) is preferably cultured under standard cell culture conditions.

(78) The arenavirus is preferably cultured in the dendritic cells or tumor cells for a period of 1 minute to 1 year, in particular 10 hours to 1 month, preferably 24 hours to 72 hours.

(79) The cultured arenavirus according to step c) is preferably isolated from a cell culture supernatant.

(80) Preferably, the dendritic cells or tumor cells are sorted according to specific properties, preferably by means of a cell sorting device, and subsequently cultured, prior to isolating according to step c). The sorted cells are preferably cultured over a period of 24 hours.

(81) In a preferred embodiment, the method further comprises the following steps: d) cloning the isolated arenavirus and e) sequencing the isolated arenavirus.

(82) With regard to further features and advantages of the method, in particular of the arenavirus and also the tumor, reference is made to the present description.

(83) According to a fourth aspect, the invention relates to a method for producing an arenavirus with tumor-regressive, i.e. tumor-repelling/counteracting, properties or improved tumor-regressive properties.

(84) The tumor is preferably a malignant tumor, preferably a carcinoma, melanoma, blastoma, lymphoma or sarcoma. Accordingly, the method is preferably a method for producing an arenavirus with (improved) carcinoma-, melanoma-, blastoma-, lymphoma- or sarcoma-regressive properties.

(85) The method comprises the following steps: a) infecting a host animal, which has a tumor, with an arenavirus, b) culturing the arenavirus in the infected host animal and c) isolating the cultured arenavirus or a subset of the cultured arenavirus from the infected host animal.

(86) The sequence of steps a) to c) may also be referred to as a (single) passage of the arenavirus in

the host animal.

(87) In a preferred embodiment, tumor tissue is transplanted to the host animal prior to carrying out step a).

(88) In an alternative embodiment, a genetically modified host animal is used which spontaneously develops the tumor.

(89) Rodents in particular, preferably mice, can be used as host animals. For example, NOD SCID mice or LoxP-Tag mice can be used as host animals. In the Nod SCID mice, the SCID (Severe Combined Immunodeficiency) mutation is combined with a NOD (non-obese diabetic) type. In mice that are homozygous for the SCID mutation, no functional T-cells or B-cells are formed. By means of this immunodeficiency, these animals are exceptionally suitable for tolerating foreign body cells, for example transplanted tumors.

(90) The host animal can be infected according to step a), for example by systemic, particularly intravenous, or local, for example subcutaneous, administration of the arenavirus.

(91) In particular, the host animal can be infected by administration, preferably injection, of the arenavirus into the tumor of the host animal.

(92) In a preferred embodiment, the sequence of steps a) to c) is repeated with a new, in particular non-infected, host animal, preferably of the same type.

(93) In a particularly preferred embodiment, the sequence of steps a) to c) is repeated many times. A new, in particular non-infected host animal, preferably of the same type, is preferably used for each repetition. In other words, it is preferred according to the invention if the arenavirus is subjected to serial passage in host animals, preferably of the same type.

(94) The sequence of steps a) to c) is preferably repeated once to 1000 times, particularly 10 times to 100 times, preferably 30 times to 60 times, wherein a new, in particular non-infected, host animal, preferably of the same type, is used for each repetition.

(95) By multiple repetition of steps a) to c), the arenavirus is constantly forced to adapt to a new environment, i.e. to a new host animal. This permanent adaptation pressure favors the occurrence of mutations, which can produce or improve the tumor-regressive properties of the arenavirus.

(96) During the culturing of the arenavirus according to step b), a replication of the arenavirus genome and a propagation of the arenavirus occur within the host animal.

(97) The arenavirus is preferably cultured in the host animal for a period of 1 minute to 500 days, in particular 10 minutes to 100 days, preferably 1 hour to 30 days.

(98) In a preferred embodiment, the arenavirus is isolated from urine, blood, the tumor, or organ lysates of the host animal.

(99) In a further embodiment, the method further comprises the following steps: d) infecting dendritic cells or tumor cells with the arenavirus isolated according to step c), e) culturing the arenavirus in the infected dendritic cells or infected tumor cells and f) isolating the cultured arenavirus or a subset of the cultured arenavirus from the infected dendritic cells or infected tumor cells.

(100) The sequence of steps d) to f) may also be referred to as a (single) passage of the arenavirus in the dendritic cells or tumor cells.

(101) The dendritic cells or tumor cells are infected according to step d) preferably by adding the arenavirus to the cells.

(102) In a preferred embodiment, the sequence of steps d) to f) is repeated with new, in particular non-infected, dendritic cells, preferably of the same type, or with new, in particular non-infected, tumor cells, preferably of the same type (the same tumor type).

(103) In a particularly preferred embodiment, the sequence of steps d) to f) is repeated many times. By way of preference, new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are used for each repetition. In other words, it is particularly preferred according to the invention if the arenavirus is additionally subjected to a serial passage in dendritic cells, preferably of the same type, or tumor

cells, preferably of the same type.

(104) The sequence of steps d) to f) is preferably repeated once to 1000 times, particularly 10 times to 100 times, preferably 30 times to 60 times, wherein new, in particular non-infected, dendritic cells, preferably of the same type, or new, in particular non-infected, tumor cells, preferably of the same type, are preferably used for each repetition.

(105) The combination of a serial passage of the arenavirus in host animals, preferably of the same type, with a serial passage of the arenavirus in dendritic cells, preferably of the same type, or tumor cells, preferably of the same type, is suitable, because of the additionally increased adaptation pressure or mutation pressure in a particular manner, for producing arenaviruses with (improved) tumor-regressive properties.

(106) The arenavirus is preferably cultured in the dendritic cells or tumor cells for a period of 1 minute to 500 days, in particular 10 minutes to 100 days, preferably 1 hour to 30 days.

(107) Preferably, the dendritic cells or tumor cells are sorted according to specific properties, preferably by means of a cell sorting device, and subsequently cultured, prior to isolating according to step f). The sorted cells are preferably cultured over a period of 24 hours.

(108) In a preferred embodiment, the method further comprises the following steps: g) cloning the arenavirus isolated according to step f) and h) sequencing the isolated arenavirus.

(109) Alternatively, it can be provided according to the invention that the arenavirus isolated according to step c) is cloned and subsequently sequenced.

(110) With regard to other features and advantages of the method, in particular the arenavirus, the tumor and also the dendritic cells and tumor cells, the present description is also fully incorporated by way of reference in order to avoid unnecessary repetitions.

(111) Further features and advantages of the invention will emerge from the following description of preferred embodiments in the form of working examples, the associated figures and the claims. The embodiments described below are merely for the purpose of illustration and for the better understanding of the invention and are in no way to be understood as limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows LCMV (WE strain) replication results in immortalized macrophages (“tumour”) and macrophages (“primary”) generated from primary bone marrow.

(2) FIG. 2 shows the effect of LCMV (WE strain) administration on MOPC (mouse plasmacytoma cell) tumor growth in C57BL/6 mice.

(3) FIG. 3 shows the effect of LCMV (WE strain) administration on MC38 (mouse colon adenocarcinoma cell) tumor growth in C57BL/6 mice.

(4) FIG. 4 shows the effect of LCMV (WE strain) administration on spontaneous liver carcinoma tumor growth in LoxP-TAg mice.

(5) FIG. 5 shows the effect of LCMV (WE strain) on interferon- γ secretion in MOPC tumor-bearing C57BL/6 mice.

(6) FIG. 6 shows the effect of LCMV (WE strain) administration on MOPC tumor growth in Map3k14.sup.aly/aly mice compared to corresponding wild type mice.

(7) FIG. 7 shows the effect of LCMV (WE strain) administration on MOPC tumor growth in Irf3 $\times$ Irf7.sup.-/- mice compared to corresponding wild type mice.

(8) FIG. 8A shows effect of LCMV (WE strain) administration on tumor microvessel density (MDV) and vessel-vessel spacing in MOPC tumor-bearing wild type mice.

(9) FIG. 8B shows effect of LCMV (WE strain) administration on tumor hypoxic regions in MOPC tumor-bearing wild type mice.

(10) FIG. 9 shows the effect of LCMV (WE strain) administration (ipsilateral, contralateral, and

intravenous) on MOPC tumor growth in C57BL/6 mice.

(11) FIG. **10** shows effect of LCMV (WE strain) administration on survival in A431 (squamous carcinoma) tumor-bearing NOD.SCID mice.

(12) FIG. **11** shows effect of LCMV (WE strain) administration on tumor growth in A431 (colon adenocarcinoma) tumor-bearing NOD.SCID mice.

(13) FIG. **12** shows effect of Candid #1 administration on tumor growth in HeLa cell-bearing NOD.SCID mice.

(14) FIG. **13** shows effect of Candid #1 administration on tumor growth in HepG2 (hepatocellular carcinoma) tumor-bearing NOD.SCID mice.

(15) FIG. **14** shows LCMV (MOI 1) replication results in primary hepatocytes, colon epithelial cells, and melanocytes) and in tumour cells from the same tissue source.

(16) FIG. **15A** shows the effect of LCMV administration on metastatic MOPC tumor growth in C57BL/6 mice.

(17) FIG. **15B** shows the effect of LCMV administration on survival in MOPC metastatic tumor-bearing C57BL/6 mice.

(18) FIG. **16** shows the effect of LCMV administration on B16F10 (melanoma) tumor growth in C57BL/6 mice.

(19) FIG. **17** shows the effect of LCMV administration on spontaneous melanoma tumor growth in MT/ret mice.

(20) FIG. **18A** shows the effect of LCMV administration on Sw872 (fibrosarcoma) tumor growth in NOD.SCID mice.

(21) FIG. **18B** shows the effect of LCMV administration on survival in Sw872 (fibrosarcoma) tumor-bearing NOD.SCID mice.

(22) FIG. **19A** shows the effect of LCMV administration on FaDu (pharyngeal carcinoma) tumor growth in NOD.SCID mice.

(23) FIG. **19B** shows the effect of LCMV administration on survival in FaDu (pharyngeal carcinoma) tumor-bearing NOD.SCID mice.

(24) FIG. **20** shows the effect of LCMV on T-cell expression of PD-1, IL2 receptor, and IL7 receptor in B16F10 tumor-bearing mice.

(25) FIG. **21A** shows the effect of LCMV administration on EL4 (lymphoma) tumor growth in C57BL/6 mice.

(26) FIG. **21B** shows the effect of LCMV administration on survival in EL4 (lymphoma) tumor-bearing C57BL/6 mice.

(27) FIG. **22** shows the effect of LCMV administration on survival in MOPC tumor-bearing C57BL/6 and Pdc1^{sup}.-/- mice.

EXPERIMENTAL SECTION

1. Methods and Materials

(28) 1.1 Mice

(29) Unless mentioned otherwise, the mice used were from a C57BL/6 background.

Map3k14^{sup.aly/aly} mice lack NF- κ B signals and are therefore highly immunosuppressed.

Irf3 $\times$ Irf7^{sup}.-/- mice cannot produce any interferon. NOD.SCID mice have no adaptive immune system. Therefore, it is possible to grow human tumors in these mice. LoxP-Tag mice spontaneously develop liver tumors.

(30) 1.2 Cell Lines and Reagents

(31) MOPC cells are murine oropharynx carcinoma cells. Mc38 are murine colon carcinoma cells. Raw cells are immortalized macrophages. A431 are human lung carcinoma cells; Sw40 are human colon carcinoma cells, Hela are human cervical carcinoma cells. Primary macrophages were cultured from bone marrow precursor cells by means of M-CSF. Cells were maintained in Dulbecco's modified Eagle's medium with 10% fetal bovine serum (Sigma-Aldrich), 2 mmol/l L-glutamine and 100 U/ml penicillin. All cells were cultured in 5% CO₂.

(32) 1.3 Viruses

(33) The LCMV strain WE was obtained from the laboratory of Prof. Zinkernagel (Experimental Immunology, Zurich, Switzerland) and was propagated in L929 cells. Candid #1 was obtained from Professor Paula Cannon (University of Southern California).

(34) 1.4 Tumor Growth and Treatments

(35) Approximately 5×10^5 tumor cells (in 100 microL) were injected subcutaneously into the right flank of 6 to 12 week old mice. The longest tumor diameter was measured by. Mice were treated by peritumoral injections of 2×10^4 PFU LCMV-WE or Candid #1 (in 100-200 microL).

(36) 1.5 Morphometric Analysis of Tumor Vessels

(37) Morphometric analyses were performed with successive frozen sections, in which the endothelial cell marker CD31 was stained. Quantification of the microvessel density (MVD) was calculated using the mean of three tumor sections. MVD was calculated as the number of vessels per tumor area.

(38) 1.6 Detection of Hypoxia

(39) Hypoxic tumor regions were detected by the formation of pimonidazole adducts after injection of pimonidazole into tumor-transplanted animals for 30 min. The tumor sections were stained using the Hypoxyprobe-1 Plus kit according to the manufacturer's instructions (Pharmacia Natur International, Inc.).

(40) 1.7 IFN- α ELISA

(41) Serum IFN- α levels were determined by ELISA according to the manufacturer's data (Research Diagnostics RDI, Flanders, NJ).

(42) 1.8 Statistical Analysis

(43) The mean values were compared using an unpaired two-sided student t-test. The data are shown as mean $\pm$ SEM. The level of statistical significance was set at $p < 0.05$.

2. Investigations

(44) 2.1 Immortalized Macrophages (Tumour Cells) and Macrophages (Primary) generated from primary bone marrow were infected with LCMV (WE strain). Replication was measured after 24 hours ($n=3$).

(45) It could be shown that LCMV (WE strain) replicates both in immortalized and healthy cells. The results obtained are shown graphically in FIG. 1. FIG. 1 has the following legend: Ordinate: LCMV (log.sub.10 PFU/ml) Abscissa: Tumor cells/healthy macrophages (primary)

(46) 2.2 WT C57BL/6 mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). Tumor growth was observed.

(47) It could be shown that the treatment with LCMV caused almost complete tumor regression. The results obtained are shown graphically in FIG. 2. FIG. 2 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)

(48) 2.3 WT C57BL/6 mice were treated with 5×10^5 MC38 cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). Tumor growth was observed.

(49) It could be shown that the treatment with LCMV caused a significant tumor regression. The results obtained are shown graphically in FIG. 3. FIG. 3 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)

(50) 2.4 About nine month old LoxP-TAg mice with spontaneously developed liver carcinomas were infected intravenously with 2×10^6 PFU LCMV or left untreated. The tumor nodes (diameters ≥ 3 mm) were quantified macroscopically on day 6 ($n=3$) and day 20 ($n=4-5$).

(51) It could be shown that the treatment with LCMV significantly reduced the number of carcinomatous liver nodes. The results obtained are shown graphically in FIG. 4. FIG. 4 has the following legend: Ordinate: Liver nodes (number) Abscissa: Time

- (52) 2.5 WT C57BL/6 mice (n=4/group) were injected subcutaneously with 5×10^5 MOPC cells (day -3) or LCMV (WE strain) 2×10^4 PFU (day 0) or both 5×10^5 MOPC cells (day -3) and 2×10^4 PFU LCMV (day 0). Serum samples were collected on day 3 and an IFN- α -ELISA was performed.
- (53) It could be shown that the LCMV caused a drastically increased secretion of interferon- γ in experimental animals which were simultaneously administered carcinoma cells. The results obtained are shown graphically in FIG. 5. FIG. 5 has the following legend: Ordinate: IFN- α (pg/ml)
- (54) 2.6 Map3k14^{sup.aly/aly} mice and WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). Tumor growth was observed.
- (55) It could be shown that the treatment with LCMV caused tumor regression. The results obtained are shown graphically in FIG. 6. FIG. 6 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)
- (56) 2.7 Irf3 $\times$ Irf7^{sup.-/-} mice and WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). Tumor growth was observed.
- (57) It could be shown that the treatment with LCMV caused tumor regression. The results obtained are shown graphically in FIG. 7. FIG. 7 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)
- (58) 2.8 WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). On day 9 after the tumor graft, the tumors were analyzed histologically with CD31 staining. The microvessel density (MDV) and the vessel-vessel spacing were quantified.
- (59) It could be shown that the treatment with LCMV caused a decrease in tumor vessel density. The results obtained are shown graphically in FIG. 8A. FIG. 8A has the following legends: Left side: Right side: Ordinate: MVD [mm^{sup.3}] Ordinate: Vessel-vessel spacing (μ m) Abscissa: Tumor/Tumor LCMV Abscissa: Tumor/Tumor LCMV
- (60) 2.9 WT mice were treated with 5×10^5 MOPC cells (day 3). One group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 0). On day 9, the animals were injected with pimonidazole, and the tumors were then analyzed histologically for hypoxic regions.
- (61) It could be shown that the treatment with LCMV caused oxygen deficiency in the carcinoma tissue. The results obtained are shown graphically in FIG. 8B. FIG. 8B has the following legend: Ordinate: Hypoxic regions/tumor (%) Abscissa: Tumor/Tumor LCMV
- (62) 2.10 WT C57BL/6 mice were injected subcutaneously with 5×10^5 MOPC cells in the right flank (day 3). On day 0, a group of animals were treated with 2×10^4 PFU LCMV (WE strain) in the right flank (ipsilateral), left flank (contralateral) or intravenously. Tumor growth was observed.
- (63) It could be shown that the treatment with LCMV caused tumor regression even with systemic administration. The results obtained are shown graphically in FIG. 9. FIG. 9 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)
- (64) 2.11 NOD.SCID mice were injected subcutaneously with 5×10^5 A431 cells (day -3) and then either left untreated or treated with 2×10^4 PFU LCMV (WE strain). The tumor size (longest diameter) was measured on the specified day. The mice were sacrificed when the tumor size reached 12 mm.
- (65) It could be shown that the treatment with LCMV increased the survival rate in the experimental animals. The results obtained are shown graphically in FIG. 10. FIG. 10 has the following legend: Ordinate: Survival (%) Abscissa: Time (days)
- (66) 2.12 NOD.SCID mice were treated with 5×10^5 Sw40 cells (day 0). A group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) or 2×10^4 PFU

Candid #1 (day 0). Tumor growth was observed.

(67) It could be shown that the treatment with LCMV and Candid #1 caused tumor regression. The results obtained are shown graphically in FIG. 11. FIG. 11 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)

(68) 2.13 NOD.SCID mice were treated with 5×10^5 Hela cells (day 0). A group of mice was additionally treated peritumorally with 2×10^4 PFU LCMV (WE strain) (day 3). Tumor growth was observed.

(69) It could be shown that the treatment with LCMV caused tumor regression. The results obtained are shown graphically in FIG. 12. FIG. 12 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)

(70) 2.14 NOD.SCID mice were treated with 5×10^5 HepG2 cells (day 10) and then additionally treated peritumorally with or without 2×10^4 PFU Candid #1 (day 0). Tumor growth was observed.

(71) It could be shown that the treatment with Candid #1 caused tumor regression with this tumor type. The results obtained are shown graphically in FIG. 13. FIG. 13 has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)

(72) 2.15 Primary human cells (hepatocytes, colon epithelial cells, melanocytes) and tumour cells from the same tissue source were infected with LCMV (MOI 1). The amount of virus was measured in the supernatant after 1, 2 and 3 days.

(73) In this experiment it was shown that arenaviruses are replicated in tumor cells in comparison to healthy tissue. FIG. 14 has the following legend: Ordinate: Infectious virus in cell culture supernatant (logarithmic plaque forming units) Abscissa: Time (days)

(74) 2.16 Tumor diameter (A) and survival (B) of C57BL/6 mice bearing a metastasis in the shoulder and a metastasis in the flank (MOPC cells), which were left untreated or had been treated intravenously with 2×10^6 PFU LCMV.

(75) It could be shown in this experiment that intravenous therapy of LCMV acts very efficiently on two local metastases and thus prolongs survival. FIG. 15A has the following legend: Ordinate: Tumor diameter of both metastases (cm) Abscissa: Time (days)

FIG. 15B has the following legend: Ordinate: Survival in percent Abscissa: Time (days)

(76) 2.17 Tumor diameter of C57BL/6 mice bearing a melanoma (B16F10 cells) which were left untreated or were treated intratumorally with 2×10^4 PFU LCMV.

(77) It could be shown in this experiment that local therapy with LCMV is very efficient in melanoma. FIG. 16 has the following legend: Ordinate: Tumor diameter of the melanoma (cm) Abscissa: Time (days)

(78) 2.18 Number of melanomas of MT/ret mice (develop endogenous melanomas), which were left untreated or were treated intravenously with 2×10^6 PFU LCMV.

(79) It could be shown in this experiment that systemic therapy with LCMV is very efficient in melanoma. FIG. 17 has the following legend: Ordinate: Number of melanomas

(80) 2.19 Tumor diameter (A) and survival (B) of NOD.SCID mice bearing a human fibrosarcoma (Sw872 cells), which were left untreated or were treated intratumorally with 2×10^6 PFU Candid #1.

(81) It could be shown in this experiment that Candid #1 acts very efficiently also in the case of fibrosarcoma and thus prolongs survival. FIG. 18A has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)

FIG. 18B has the following legend: Ordinate: Survival in percent Abscissa: Time (days)

(82) 2.20 Tumor diameter (A) and survival (B) of NOD.SCID mice bearing a human pharyngeal carcinoma (FaDu cells), which were left untreated or were treated intratumorally with 2×10^6 PFU LCMV.

(83) It could be shown in this experiment that LCMV acts very efficiently also in the case of pharyngeal carcinoma and thus prolongs survival. FIG. 19A has the following legend: Ordinate:

Tumor diameter (cm) Abscissa: Time (days)

FIG. 19B has the following legend: Ordinate: Survival in percent Abscissa: Time (days)

(84) 2.21 Expression of receptors on tumor-specific T cells (PD-1, IL2 receptor, IL7 receptor), which influence the function of T cells. Tumor-specific T cells are derived from the blood of mice with B16F10 tumors, which were additionally treated intratumorally with or without LCMV.

(85) It could be shown in this experiment that LCMV positively influences the tumor-specific T cells. FIG. 20 has the following legend: Ordinate: Potency of the expression of the different receptors (mean fluorescence intensity)

(86) 2.22 Tumor diameter (A) and survival (B) of C57BL/6 mice bearing a murine subcutaneous lymphoma (EL4 cells) which were treated with or without tumor-specific T cells (OT1 cells) and additionally intratumorally with or without LCMV (2×10^6 PFU).

(87) It could be shown in this experiment that LCMV acts synergistically with T cell therapy. FIG. 21A has the following legend: Ordinate: Tumor diameter (cm) Abscissa: Time (days)

FIG. 21B has the following legend: Ordinate: Survival in percent Abscissa: Time (days)

(88) 2.23 Survival of C57BL/6 mice and PD-1 deficient mice (Pdc1^{sup.-/-} mice) bearing a murine pharyngeal carcinoma (MOPC cells) and which were treated intratumorally with LCMV (2×10^4 PFU).

(89) It could be shown in this experiment that LCMV acts synergistically with a PD-1 blockade.

FIG. 22 has the following legend: Ordinate: Survival in percent Abscissa: Time (days)

(90) The nucleic acid sequences SEQ ID No. 1 to SEQ ID No. 6 mentioned in the general description correspond to the nucleic acid sequences disclosed in the following sequence listing.

Claims

1. A method of treating a tumor comprising administering to a subject in need thereof a lymphocytic choriomeningitis virus (LCMV), wherein the sequences of the L-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile L-ribonucleic acid sequences, and the S-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile S-ribonucleic acid sequences.
2. The method of claim 1, wherein the method comprises activating innate (congenital) or adaptive immune cells and thereby effecting tumor regression.
3. The method of claim 1, wherein the method comprises effecting increased secretion of anti-tumoral cytokines by immune cells by activating said immune cells.
4. The method of claim 3, wherein the anti-tumoral cytokine is interferon- α , interferon- β or interferon- γ .
5. The method of claim 1, wherein the tumor is selected from the group consisting of carcinoma, melanoma, blastoma, lymphoma, and sarcoma.
6. The method of claim 1, wherein the arenavirus comprises a nucleic acid comprising an S-ribonucleic acid sequence according to SEQ ID No. 5 or antisense sequence thereof, or a nucleic acid comprising an L-ribonucleic acid sequence according to SEQ ID No. 6 or antisense sequence thereof.
7. The method of claim 1, wherein the lymphocytic choriomeningitis virus is a lymphocytic choriomeningitis virus strain WE.
8. The method of claim 1, wherein the subject is a human.
9. The method of claim 1, wherein the L- and S-ribonucleic acids of the LCMV are WE L- and S-ribonucleic acids.
10. A method of treating a tumor comprising administering to a subject in need thereof a lymphocytic choriomeningitis virus (LCMV), wherein the sequences of the L-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile L-ribonucleic acid sequences, the S-ribonucleic acids of the LCMV are selected from the group consisting of WE,

Clone 13 and Docile S-ribonucleic acid sequences, and the tumor is a solid tumor.

11. The method of claim 10, wherein the tumor is carcinoma.

12. The method of claim 10, wherein the tumor is melanoma.

13. The method of claim 10, wherein the subject is a human.

14. A method of treating a tumor comprising administering to a subject in need thereof a lymphocytic choriomeningitis virus (LCMV), wherein the sequences of the L-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile L-ribonucleic acid sequences, the S-ribonucleic acids of the LCMV are selected from the group consisting of WE, Clone 13 and Docile S-ribonucleic acid sequences, and the tumor is selected from the group consisting of oropharynx cancer, colon cancer, lung cancer, cervical cancer, liver cancer, melanoma, pharyngeal cancer, lymphoma, thyroid cancer, and bladder cancer.

15. The method of claim 14, wherein the tumor is selected from the group consisting of oropharynx cancer, colon cancer, lung cancer, cervical cancer, liver cancer, melanoma, pharyngeal cancer, and lymphoma.

16. The method of claim 14, wherein the subject is a human.
